MR. ALLEN'S MATH CLASS

Right Triangle Treasure Map Project

Unit 5 · Pythagorean Theorem · NC.8.G.6–8

Name: _____ **Class Period:** _____ **Due Date:** _____

The Mission

You are a treasure-map maker! Design a treasure map on a coordinate grid where explorers must travel between landmarks. Every path between landmarks is the hypotenuse of a right triangle — and explorers need YOU to calculate each distance using the Pythagorean Theorem: $a^2 + b^2 = c^2$.

Requirements

- **The Map (20 pts):** Draw a coordinate grid (at least 20 × 20). Place and label **5 landmarks** at points with whole-number coordinates (example: Skull Rock at (3, 14)). Make it colorful and creative!
- **The Paths (40 pts):** Connect the landmarks with **4 straight paths** in order (a treasure trail). No path may be perfectly horizontal or vertical — every path must be a true diagonal.
- **The Math (30 pts):** For each path, draw the right triangle (dashed legs), label the legs a and b, then show every step of the Pythagorean Theorem to find the length. Round to the nearest tenth.
- **The Legend (10 pts):** Include a story caption (2–3 sentences), a compass rose, and the **total length of the treasure trail** (sum of all 4 paths).

Show-Your-Work Table

Path 1	From (____ , ____) to (____ , ____)	a = _____ b = _____	c = _____
Path 2	From (____ , ____) to (____ , ____)	a = _____ b = _____	c = _____
Path 3	From (____ , ____) to (____ , ____)	a = _____ b = _____	c = _____
Path 4	From (____ , ____) to (____ , ____)	a = _____ b = _____	c = _____

Grading Rubric

Category	4 — Excellent	3 — Good	2 — Developing	1 — Beginning
Map & landmarks	5 labeled landmarks, neat grid, creative theme	5 landmarks, readable grid	3–4 landmarks or messy grid	Fewer than 3 landmarks
Right triangles	All 4 triangles drawn with legs labeled	3 triangles correct	2 triangles correct	0–1 triangles correct

Calculations	All 4 distances correct with all steps shown	3 correct with steps	2 correct or steps missing	0–1 correct
Legend & total	Story, compass, and correct total	Missing one element	Missing two elements	Missing most elements

Bonus (+5): Add a 5th path and use the distance formula version of the theorem to check one of your answers.